

Coverage Across the Horizon

Horizon- and Channel-Conditional Adaptive Conformal Calibration
for Long-Term Time-Series Forecasting

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Draft v2 — 1 September 2026

Code, data scripts and every result artefact:

<https://github.com/aarush093/coverage-across-the-horizon>

Abstract

Long-term time-series forecasting (LTSF) is scored by point error, and the conformal methods now being attached to LTSF forecasters are mostly scored the way point forecasts are: by one aggregate coverage number. We show that this hides the failure that matters. Auditing conditional coverage over a horizon-bucket $\times$ channel grid on the four ETT benchmarks, the per-horizon split-conformal baseline (MSCP) has the *highest* marginal coverage of seven interval methods (0.9123) and the *lowest* worst-cell coverage (0.7251), below a global split-conformal quantile (0.7667). The obvious repairs do not work on their own: online adaptation alone reliably lowers worst-cell coverage (0.7446); horizon $\times$ channel conditioning alone is inconsistent (0.7588, better than global on three of four datasets and much worse on the fourth). Their combination reaches 0.8591 with intervals no wider than the global baseline, an interaction of +0.1224 (bootstrap 95% CI (+0.088, +0.163)) that holds in 31/32 configurations and under an honest online protocol in which every feedback signal is used only once realised. On a 300-cell electricity-load surface with a seasonal calibration-test shift, static conditioning collapses to 0.4621 and the combination reaches 0.8059 (interaction +0.3022). The layer is post-hoc, distribution-free, model-agnostic and trains nothing. We also report the width price of whole-path coverage on both surfaces ($3.25\times$ on ETT, $10.16\times$ on electricity), a decision-level evaluation in which interval gating pays only when misses cost $\gtrsim 5\times$ false alarms, and two methodological artefacts—a scoring confound and an online-feedback leak—that we found in our own pipeline and report in full.

1 Introduction

The modern LTSF literature is organised around a shared evaluation contract: deterministic point forecasts, scored by MSE and MAE, on a fixed benchmark suite, at horizons of 96 to 720 steps [1–3]. The three papers that define the contract disagree about architecture almost completely and agree, silently, that no uncertainty need be reported. Because they disagree on everything else, the omission is a property of the contract rather than of any one paper. The ProbtS benchmark documents the resulting split: long-horizon point forecasting and short-horizon probabilistic estimation developed as separate literatures [4].

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Conformal prediction is the natural repair. It is distribution-free, post-hoc, and requires no retraining, which matters when the forecaster is a frozen artefact—the normal deployment condition. A 2021–2026 line attaches conformal methods to forecasting: adaptive conformal inference (ACI) for online coverage [5, 6], conformal PID control [8], autocorrelation-aware multi-step calibration [9], and joint multi-step constructions [11–13].

The problem this paper addresses. That line has largely inherited the point literature’s scoring habit. The closest benchmarking work to ours evaluates conformal methods for time series on marginal coverage, interval width and the Winkler score, with AutoARIMA on a monthly-sales corpus, and concludes that multi-step split conformal prediction (MSCP) is the best available method [26]. Under the same three metrics, MSCP is the best-covered and a competitive method on our surface too—and under conditional scoring it is the worst of seven. A method can hold 90% coverage on average while failing systematically at long horizons, on particular channels, or during particular seasons. Averages hide exactly the failures that make an interval worth having.

Contributions.

1. **An audit.** A conditional-coverage map over a horizon-bucket $\times$ channel grid on the four ETT benchmarks and a screened Electricity subset, across seven interval methods and two frozen backbones (Section 5). Marginal and conditional coverage rank the methods in opposite orders.
2. **A method, and the finding that its two halves are inseparable.** A bucketed horizon $\times$ channel conditional adaptive conformal layer (Section 4). Adaptation alone reliably hurts worst-cell coverage; conditioning alone is inconsistent; their combination is worth $+0.0924$ over the global baseline on ETT and $+0.2302$ on electricity load, and the same interaction appears within the conditioning, on the horizon axis specifically (Sections 6–7).
3. **The price of whole-path guarantees.** Per-step 90% intervals give essentially zero probability that an entire trajectory is covered. Restoring ≈ 0.9 whole-path coverage costs $3.25\text{--}3.59\times$ the width on ETT and $\approx 10.16\times$ on electricity (Section 9).
4. **A decision-level evaluation, reported in both directions.** On interval-gated peak-load flagging, calibrated intervals beat a point-forecast rule only when misses cost at least $\approx 5\times$ false alarms; below that the point rule is cheaper (Section 10).
5. **Two negative methodological results.** Scoring a bucket-count ablation on each setting’s own grid reverses its conclusion (Section 7); and an online update that consumes a path’s full-horizon outcome before the next path is issued uses up to $H - \text{stride}$ steps of future information while passing a standard leakage test (Section 3). Both bit us before they became findings; both are cheap to check.

2 Related work and positioning

Conformal prediction for time series. Split conformal prediction [21, 22] assumes exchangeability, which time series violate. ACI updates the target level online from realised coverage and guarantees long-run coverage without distributional assumptions [5]; DtACI and AgACI remove the fixed step size [6, 7]; conformal PID control generalises the update [8]. Multi-step variants calibrate per horizon (MSCP) or model cross-horizon error dependence with the feedback delay that

multi-step forecasting imposes [9, 10]. Joint constructions target whole-trajectory coverage through Bonferroni correction [11], dynamic programming [12] or copulas [13]. Exact distribution-free conditional coverage is impossible in general [17–19]; approximate and group-conditional guarantees exist [20].

What the two halves are. Our static conditional arm is Mondrian conformal prediction [14] with the taxonomy fixed to (horizon bucket, channel). Our adaptive arm runs one ACI tracker per Mondrian cell, which is the group-conditional online setting studied, with stronger guarantees than we claim, by multivalid conformal prediction [15] and, concurrently, by online localised conformal prediction with covariate-kernel localisation [16]. We are therefore not the first to condition, not the first to adapt, and not the first to do both. The contribution is (i) that on the LTSF grid, under conditional scoring, each half on its own is neutral or harmful; (ii) that the combination is what works, which to our knowledge no prior work reports on LTSF backbones at this scale; and (iii) the audit and decision layer around it.

Nearest neighbours. Sabashvili [26] is the closest benchmarking work and our central foil (Section 5); it overlaps our setting in neither data, backbone nor metric. BC-ACI [27] is the closest method on the adaptation axis: it estimates forecast bias online per horizon and re-centres the interval. It moves the interval’s centre; we condition its width; it has no channel dimension. We measure its premise on our surface in Section 11.1. DeRegiME [28] is the nearest work by data and backbone—probabilistic forecasting on the same suite with DLinear and PatchTST encoders—but trains a sparse-GP density model optimising NLPD and CRPS, where we perform distribution-free post-hoc calibration with coverage control and train nothing. ABF-T-GLCP [29] localises calibration residuals by a learned gate state and recency, i.e. a data-driven taxonomy attached to its own forecasting module, and claims approximate local coverage under stability conditions; we use a fixed taxonomy, adapt online, and claim only long-run adaptive coverage. Architectural work on cross-channel and exogenous modelling [30, 31] is out of scope: we contribute calibration, not architecture.

3 Setup

Backbones. We use DLinear and NLinear [2] as frozen point forecasters with look-back $L = 336$, fitted once per (dataset, horizon) by closed-form ridge least squares ($\lambda = 10^{-3}n$, bias unpenalised; DLinear kernel 25, one linear map shared across channels) and never updated. Because the objective is squared error and the models are linear, the normal equations return the global optimum of the objective that SGD approximates; there is no learning rate, epoch schedule or early stopping. This is a deviation from our written plan (Idea Lock v1.0, 16 July 2026, in the repository), which specified DLinear, PatchTST and Informer trained with the reference implementations; the two Transformer backbones were dropped under a compute-contraction rule and NLinear substituted. Section 12 states what the substitution costs. The pipeline is deterministic end to end; the published seed (2026) exists for completeness and controls nothing in the reported numbers.

Data and protocol. ETTh1, ETTh2, ETTm1, ETTm2 at $H \in \{96, 192, 336, 720\}$ form the main study (32 configurations: two backbones $\times$ four datasets $\times$ four horizons). A case study uses the LTSF Electricity benchmark [32, 33], screened to 50 of 321 meters (Section 8). Splits are the standard sequential 12/4/4 months (ETT) and 70/10/20 (Electricity), never random; the validation block is the calibration block. Drop-last batch truncation is disabled everywhere, since it inflates

reported LTSF scores [25]. Calibration and test windows start once per day (stride 24 on hourly data, 96 on 15-minute data). This is a deviation from the strict non-overlap rule in our plan, taken because non-overlapping windows leave 3 calibration paths at $H = 720$; at $H = 720$ adjacent daily windows still share 696 of 720 targets, so the calibration scores are strongly dependent and their count is not a sample size. Section 11 ablates the stride where sample size permits, and Section 12 carries the consequence.

Online protocol, and a leak we found. Adaptive methods update from realised coverage. In our first implementation each cell’s tracker consumed a test path’s outcomes at *all* horizon steps before the next path—one day later—was issued, so the interval for path $t+1$ at horizon 720 had already seen an outcome that in real time lies 696 steps in its future. A standard leakage check passed (perturbing the second half of the test block leaves every first-half interval bit-identical), because the leaked information lies inside the first half. Every adaptive number in this paper is under the corrected protocol: a path’s outcome in cell (k, c) reaches the tracker only once the cell’s last horizon step has been observed, a delay of $\lceil \text{edge}_k / \text{stride} \rceil$ paths (30 paths for the last bucket at $H = 720$). The original “instant-feedback” numbers are retained in Appendix A as an oracle upper bound.

Point-forecast sanity. All 32 point MSEs are in Appendix B. DLinear on ETTh1 reaches 0.3702 / 0.4041 / 0.4334 / 0.4714 at $H = 96/192/336/720$, within about 1.3% of the published values; NLinear tracks its published values similarly. DLinear on ETTh2 and ETTm2 at $H \geq 336$ is [TODO: fill from Zeng et al. Table 2, look-back 336] percent above the published figures, and ETTh2 at $H = 720$ (0.7407) sits inside a range of published reproductions that disagree among themselves. We report the whole table rather than its flattering corner; the closed-form fit lacks the implicit regularisation of Adam with early stopping and this plausibly costs a few percent at the longest horizons.

Metrics. Let $\alpha = 0.10$. For a residual tensor $R \in \mathbb{R}^{n \times H \times C}$ and half-widths $\hat{q}$, empirical coverage on cell (k, c) is the fraction of test paths and steps in horizon bucket k , channel c , with $|R| \leq \hat{q}$. We report marginal coverage; worst-cell coverage $\text{WC} = \min_{k,c}$; and, because a minimum over a wide grid is a noisy statistic, the 5th percentile of the cell distribution and the fraction of cells within ± 5 points of target. We also report normalised width, the Winkler interval score [24], whole-path (joint) coverage, and paired sign counts over configurations with paired-bootstrap confidence intervals.

Reproducibility. A clean clone, re-run end to end, reproduces the committed result file across all 12625 numeric values with maximum absolute deviation 0.0 on the reference machine; across machines the numbers agree to four decimal places. Every number in this paper, in tables and in prose, is a macro generated from the released artefacts by a script; none is transcribed.

4 Method

The layer sits entirely downstream of a frozen forecaster and is implementable in NumPy.

Nonconformity. Residuals are normalised by a per-channel scale s_c —the median absolute deviation of the calibration residuals, fixed thereafter and floored at 10^{-8} —giving scores $S = |R|/s_c$. MAD versus standard deviation is an ablation (Section 11).

Table 1: Main study: ETT $\times 4$, 42-cell grid, target 0.90, mean over 32 configurations; adaptive rows use realised feedback. “Wins” = configurations in which the method’s worst-cell coverage exceeds the global baseline’s. Gaussian conditions per (step, channel) without bucketing. Per-step intervals give essentially zero whole-path coverage at every H (Section 9).

Method	Cond. / adapt.	Marginal	Worst-cell	5th pct. cell	Within ± 5	Width	Winkler	Wins
Gaussian residual	step $\times C$ / no	0.9096	0.7762	0.8345	0.461	2.0254	2.6804	19/32
Global split CP	none / no	0.9097	0.7667	0.8255	0.226	2.0857	2.9422	—
Per-horizon (MSCP)	horizon / no	0.9123	0.7251	0.7866	0.431	2.0392	2.7649	17/32
Channel-only	channel / no	0.9099	0.7797	0.8326	0.237	2.1711	2.9806	21/32
Static conditional	$K \times C$ / no	0.9084	0.7588	0.8271	0.520	2.1304	2.9095	21/32
ACI	none / yes	0.9017	0.7446	0.8085	0.209	1.8906	2.8579	11/32
Proposed	$K \times C$ / yes	0.9069	0.8591	0.8689	0.872	1.9684	2.7100	31/32

Bucketed conditioning. Fitting H independent quantiles for $H = 720$ from a calibration block of order 10^2 paths explodes variance. We partition the horizon into K log-spaced buckets with edges $\text{unique}(\text{round}(\text{logspace}(0, \log_{10} H, K+1)))$, so that early steps get fine resolution and late steps are pooled; for $H = 720$, $K = 6$ the edges are $[1, 3, 9, 27, 80, 240, 720]$, and at small H rounding can yield fewer than K buckets (9 at $H = 96$, $K = 10$). Conditioning is on the product grid (bucket $\times$ channel): $K \times C$ cells, 42 on ETT and 300 on the electricity surface. K is an ablation (Section 7).

Split-conformal quantile. Within each cell we take the $\lceil (n+1)(1-\alpha) \rceil / n$ empirical quantile of the cell’s calibration scores, upper-interpolated. Since the scores come from overlapping windows, n counts dependent observations and the finite-sample form is a convention, not a guarantee.

Online adaptation. Each cell carries its own ACI-style tracker. Once a path’s outcome in the cell is realised (Section 3), the cell’s working level is updated with the path’s empirical miss fraction over the cell’s steps, $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{miss}_t)$, one update per path per cell. The running level is unclamped, which preserves the ACI memory; the effective level is clipped to $[10^{-4}, 0.5]$ when the quantile is read from the cell’s own calibration scores, so an interval can never exceed the cell’s largest calibration score. We fix $\gamma = 0.02$ before scoring any test block and use the same γ for the unconditional ACI baseline. The tracker is per-cell: this is the only structural difference from ACI, and Section 6 shows it is the difference that matters.

Cost. All seven methods on the largest ETT tensor ($91 \times 720 \times 7$ test intervals) run in **[TODO: re-time on the reference machine and inject]** on a laptop CPU; the calibration layer is negligible beside any forecaster.

Theory scope. We claim only what the construction supports: ACI-family long-run adaptive coverage per cell, plus approximate conditioning at bucket resolution. We do not claim finite-sample validity under distribution shift, and we do not claim exact conditional coverage, which is impossible distribution-free [17, 19]. Bucketing is a variance-control device, not a guarantee.

5 The audit: marginal coverage is uninformative

Table 1 compares seven interval methods spanning the 2×2 of conditioning and adaptation, plus a Gaussian-residual and an MSCP baseline, under the realised-feedback protocol.

Read the marginal column alone and the seven methods are indistinguishable: every one lands within a point of the 90% target. Read the worst-cell column and they span fourteen points; read the within- ± 5 column and the global baseline has fewer than a quarter of its cells near target while the proposed layer has 0.872.

The ranking inverts. MSCP has the *best* marginal coverage in the table and the *worst* conditional coverage—0.7251, four points below simply using one global quantile. This is not a quirk of our implementation: a benchmark scoring marginal coverage, width and Winkler, as Sabashvili [26] does, would rank MSCP first on coverage and never see that it is the method that fails hardest per cell. Per-horizon conditioning spends a fixed calibration budget across H separate quantiles, so each is estimated from fewer effective samples, and the variance shows up in the minimum; Section 8.2 confirms that on ETT this failure shrinks with calibration data (MSCP gains +0.176 of worst-cell across a fourfold increase), i.e. it is variance, not structure.

6 Results: the interaction is the finding

The four rows of the 2×2 in Table 1 are the experiment:

$$\begin{array}{cccc} \underbrace{0.7667} & \underbrace{0.7588} & \underbrace{0.7446} & \underbrace{0.8591} \\ \text{neither} & \text{conditioning only} & \text{adaptation only} & \text{both} \end{array}$$

Neither intervention helps on its own, and the two fail differently. Adaptation alone is reliably worse than the do-nothing baseline: -0.0221 on average, better in only 11/32 configurations, 95% CI (-0.032, -0.012). Conditioning alone is inconsistent: -0.0079 on average with a CI (-0.042, +0.024) that spans zero, better than global in 21/32 configurations and on three of the four datasets (ETTh1 0.8611 vs. 0.8008), but far worse on ETTh1 (0.6085 vs. 0.7680), where the 42 static cells are estimated from too little data. The combination is not inconsistent: +0.0924 over global, CI (+0.076, +0.109), in 31/32 configurations. The interaction term is

$$P - C - A + G = +0.1224 \quad (95\% \text{ CI } (+0.088, +0.163)),$$

and it is positive on the grid-size-invariant statistics too (5th-percentile cell +0.0588, within- ± 5 fraction +0.3691).

The mechanism is intelligible. Conditioning alone slices a finite calibration block into 42 cells and each cell’s static quantile inherits the variance—concentrated in the minimum, and on ETTh1 catastrophically. Adaptation alone has enough data but one global knob, so it cannot move a cell that is failing while the aggregate looks healthy; it purchases marginal coverage by narrowing (width 1.8906, the narrowest in the table) and its worst cell degrades. Together, conditioning supplies the resolution to see per-cell failure and adaptation supplies the correction.

Width and score. Proposed is narrower than the global baseline (1.9684 vs. 2.0857, ratio 0.97) and has the best Winkler score of the six conformal methods (2.7100). The parametric Gaussian baseline is narrowly ahead on Winkler (2.6804) while sitting nine points below on worst-cell coverage; under the instant-feedback oracle (Appendix A) Proposed led on Winkler outright, and the difference is the price of realised feedback at long horizons.

Consistency. Against MSCP and ACI, Proposed wins on worst-cell in 32/32 and 32/32 configurations; against static conditional 31/32, channel-only 30/32, global 31/32, and the Gaussian baseline 26/32. The gain over global is present at every horizon (+0.1037, +0.1212, +0.0839, +0.0610 at $H = 96, 192, 336, 720$) and smallest at $H = 720$, where the last bucket’s tracker waits 30 paths for its first realised update.

Backbone swap. The layer never sees the forecaster. Swapping DLinear for NLinear, Proposed reaches worst-cell 0.8604 and 0.8579 respectively, while the global baseline moves by three points (0.7510 vs. 0.7825). The two backbones have materially different error processes (Section 11.1), which makes the agreement more informative than two backbones that merely share a name—but they are two members of one linear family, and Section 12 does not call this independence.

7 Both halves of the conditioning need the adaptation

Setting $K = 1$ collapses the horizon axis, leaving channel-only conditioning; we verified structurally that Proposed at $K = 1$ equals the channel-only adaptive arm exactly in all 32 configurations. Sweeping K and scoring every setting on the *same* fixed $K = 6$ grid (instant feedback; the ablation predates the protocol change and is **[TODO: re-run under realised feedback]**):

Surface	Arm	$K=1$ (channel only)	$K=6$ (horizon $\times$ channel)	Horizon-axis effect
ETT	Static (Cond)	0.7797	0.7588	-0.0209
ETT	Adaptive (Proposed)	0.8263	0.8657	+0.0394
Electricity	Static (Cond)	0.5473	0.4621	-0.0853
Electricity	Adaptive (Proposed)	0.8340	0.8306	-0.0034

On ETT the horizon axis *hurts* without adaptation and *helps* with it: a second interaction of +0.0603, with $K = 6$ beating $K = 1$ in 31/32 configurations on the adaptive arm. On electricity the axis hurts the static arm badly (-0.0853) and adaptation *neutralises* the harm rather than converting it into a gain (-0.0034); the interaction (+0.0818) is carried by the static collapse. The honest statement is that the horizon axis is safe under adaptation and unsafe without it, and pays on ETT.

A scoring artefact that reversed this conclusion. Our first K ablation scored each K on *its own* grid. Worst-cell coverage is a minimum over $K \times C$ cells—7 cells at $K = 1$, 70 at $K = 10$ —and a minimum over fewer, larger cells is higher almost by construction. Own-grid scoring therefore decreases monotonically in K (0.8717 at $K=1$ to 0.8616 at $K=10$) and recommends discarding the horizon axis; fixed-grid scoring recommends $K = 6$. The sign test flips completely: $K = 6$ beats $K = 1$ in 31/32 configurations on a fixed grid and 0/32 on own grids. The two agree exactly at $K = 6$, where the grids coincide, which is the consistency check that nothing else moved.

We report this because the confounded figure was presented before the error was found, and because the rule is reusable: **never compare a minimum or maximum across settings that change the number of cells it is taken over.** The same rule forbids ranking our two surfaces by their interactions: a minimum over 300 cells is not a minimum over 42, and where we compare surfaces below we use the distributional statistics instead.

Table 2: Electricity, 50 meters, 300-cell grid, target 0.90, mean over 8 configurations (two backbones $\times$ four horizons); adaptive rows use realised feedback.

Method	Marginal	Worst-cell	5th pct. cell	Within ± 5 pt	Width	Winkler
Gaussian residual	0.8330	0.4270	0.6314	0.500	0.8503	1.6702
Global split CP	0.8644	0.5757	0.7995	0.534	0.8568	1.5263
Per-horizon (MSCP)	0.8472	0.5382	0.7158	0.511	0.8402	1.5925
Channel-only	0.8629	0.5473	0.7860	0.525	0.8729	1.5456
Static conditional	0.8610	0.4621	0.6725	0.522	0.8702	1.5417
ACI	0.8906	0.6173	0.8340	0.493	0.9587	1.5251
Proposed	0.8931	0.8059	0.8768	0.972	1.0339	1.5301

8 Case study: electricity load under seasonal shift

We screen the 321-client LTSF Electricity benchmark to meters with no zero readings—a measured criterion, not an assumed one: 92 of 321 qualify and we keep the first 50 by column order, yielding a 300-cell grid. The sequential split places calibration in late winter and spring and the test block from late spring through the end of the year, so the evaluation crosses a genuine seasonal transition. Because a minimum over 300 cells is a noisy statistic, Table 2 reports the grid distribution alongside it.

Under shift every static method under-covers even marginally, so here the marginal column is informative too. The pattern of Section 6 sharpens: static conditioning is actively harmful (0.4621, eleven points below the global baseline); adaptation alone recovers part of it (0.6173); the combination reaches 0.8059, with 0.972 of the 300 cells within five points of target against 0.534 under global conformal. The interaction is $+0.3022$ on the minimum and $+0.1698$ on the 5th-percentile cell. The gain is not free on this surface: under realised feedback Proposed is $1.20\times$ the width of the global baseline and its Winkler score (1.5301) is on par with global and ACI rather than ahead of them. Conditional coverage under shift is bought with width here; on ETT it was not.

8.1 Adaptation, observed rather than averaged

A block average cannot distinguish a method that holds target from one that over-covers and then collapses. Figure 1 plots rolling coverage through the test block. **[TODO: regenerate both figures from the realised-feedback artefacts; the numbers in this section are realised, the plotted curves are not yet]**

Across the eight electricity configurations, global conformal spends 17.5% of the test period below 0.85 coverage (five points under target, the half-width of the “within ± 5 ” band), with excursions averaging 21.8 days. The adaptive methods spend about a third of that—6.5% for ACI and 6.3% for Proposed—and recover faster, with excursions averaging 8.2 and 9.0 days; their worst rolling window is three points higher (0.8412 against 0.8147). Adaptation shortens and shallows the seasonal failure; it does not abolish it. Under our earlier oracle-feedback protocol the adaptive methods spent no time at all below 0.85, and that figure was an artefact of consuming outcomes before they were observed (Section 3). The static failure is not diffuse: its worst window falls at the same test day in seven of eight configurations and for both backbones, i.e. it is a property of the data, and it deepens with horizon.

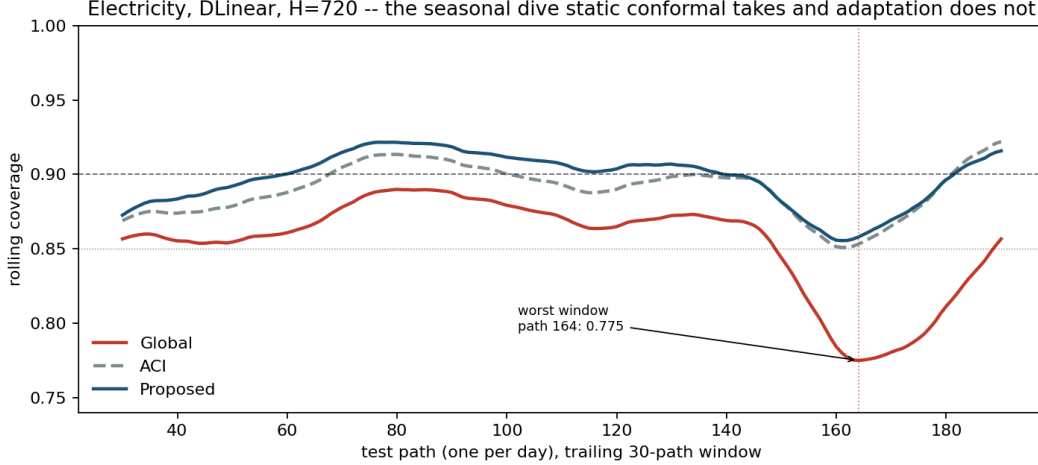


Figure 1: Rolling 30-day coverage, Electricity, DLinear, $H = 720$. Static global conformal dips through the autumn transition to 0.7750 at test day 164; the conditional adaptive layer holds 0.8052 through the identical window on identical forecasts.

Reported against ourselves. On ETT, whose split does not cross a season, the effect disappears: global spends 16.0% of the period below 0.85 and Proposed 16.3%, which is no better, while ACI alone is *worse than doing nothing* (22.8%). What survives on ETT is the shape of the failure rather than its frequency: Proposed’s dips are shorter (9.4 days against 13.2) and its worst window no deeper (0.8592 against 0.8585). Rolling-coverage stability is therefore a shift phenomenon—worth something when the test block crosses a season, worth nothing when it does not. Adaptation is not a free win, and we would rather report that than the surface on which it looks like one.

8.2 How much calibration data does this need?

Truncating the calibration block from the front (keeping the most recent paths) separates a structural limitation from a statistical one, and the answer differs by surface. On ETT every method is data-limited, and the method that gains most is the one that fails hardest: across a fourfold increase in calibration paths (28 to 112) MSCP gains +0.176 of worst-cell—more than Proposed’s +0.125—while global gains +0.109 and static conditional +0.139. That is the direct evidence for the variance mechanism of Section 5: on ETT the baselines are short of data, not structurally broken. Proposed still leads at every size, including at $n_{\text{cal}} = 28$ (0.7343 against the best baseline’s 0.6676). On electricity the two failure modes separate cleanly: the global baseline gains +0.034 and MSCP +0.008, while Proposed gains +0.175 (0.6307 to 0.8059) and has not saturated (Figure 2). Under the seasonal shift the baselines’ failure is structural—more data does not help them—whereas ours is statistical: it is estimating a 300-cell grid from fewer than a hundred paths. The reported electricity numbers are therefore a floor, and the honest limitation is that wide grids need data.

9 Whole-path coverage and its price

Per-cell 90% coverage is not 90% coverage of an entire trajectory, and conflating the two would be a serious overclaim. We measure both. Every per-step method achieves essentially zero joint coverage: on ETT the probability that all steps across all channels are simultaneously covered is 0.0021 for Proposed and 0.0158 for the Gaussian baseline at $H = 96$, and 0.0000 for every method at $H \geq 336$.

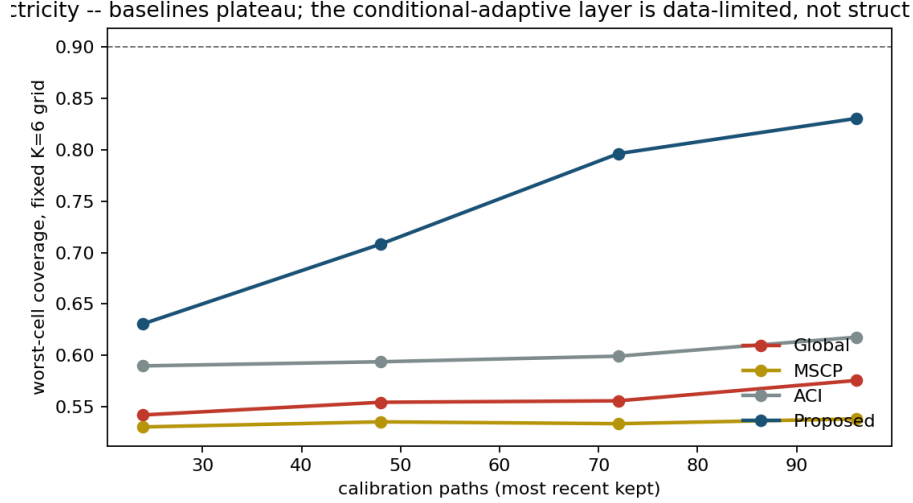


Figure 2: Worst-cell coverage against calibration size, Electricity, fixed $K = 6$ grid. Baselines plateau; the conditional adaptive layer is data-limited, not structure-limited. **[TODO: regenerate from the realised-feedback artefact; add the static-conditional curve; remove the in-plot title]**

A max-normalised-score layer [11, 13] on ETT restores whole-path coverage to 0.8772 at $3.25\times$ the width (per H : 0.8916, 0.9018, 0.8825, 0.8330); a Bonferroni layer gives 0.9154 at $3.59\times$. On electricity the same layers reach 0.9124 and 0.9130 at $10.16\times$ and $10.19\times$. Whole-path coverage on the LTSF grid is available and its price ranges from three to ten times the interval width depending on the surface. That is a finding about the regime, not a defect of any method, and it is why we report both numbers.

Joint coverage is also strongly finite-sample. On electricity, holding calibration size fixed at 80 paths, whole-path coverage is 0.9259 at $H = 96$ against 0.8947 at $H = 720$, while at full calibration $H = 96$ reaches 0.9421; the decay from $H = 96$ to $H = 720$ therefore decomposes into $(0.9259 - 0.8947)/(0.9421 - 0.8947) \approx 63\%$ horizon effect and 37% sample-size effect **[TODO: inject these four figures from cw_ecl.json]**. At $H = 720$, halving calibration from 80 to 40 paths drops joint coverage from 0.8947 to 0.5632.

10 Decision-level evaluation

Coverage is instrumental. On the electricity surface (mean over 8 configurations, all four horizons) we evaluate interval-gated peak-load flagging: a per-meter peak threshold is set at the 0.95 quantile of the calibration block only (we verified by source inspection and by perturbation that no test data enters it), and a meter-hour is flagged when the point forecast, or the upper interval endpoint, exceeds it. Costs are normalised so that flagging nothing costs 1.0.

Miss : false-alarm	Flag all	Point forecast	Interval (global)	Interval (Proposed)
2:1	2.5140	0.4625	0.6703	0.6919
5:1	1.0056	0.3872	0.2988	0.2981
10:1	0.5028	0.3621	0.1749	0.1668
20:1	0.2514	0.3495	0.1130	0.1012
50:1	0.1006	0.3420	0.0758	0.0618

Calibrated intervals are not universally better. At 2:1 the point rule is cheaper, because interval gating flags aggressively (recall 0.96 at precision 0.42, against 0.66 at 0.73 for the point rule) and pays for it in false alarms. From roughly 5:1 the ordering reverses and widens: at 10:1 the proposed layer costs 0.1668 against the point rule’s 0.3621. On the worst channel the picture is worse: at 2:1 the interval rules reach $8.36\text{--}10.23\times$ the do-nothing cost. Two caveats bound the claim. The point and interval rules operate at different points on the recall–precision curve, and a point rule with a constant margin tuned on the calibration block is the fair comparator ([**TODO: tuned-margin baseline**]); until it is reported, the result says that a conformal margin beats *no* margin in asymmetric-cost regimes, not that it beats every margin. And the decision numbers were produced with instant feedback ([**TODO: re-run**]). Interval gating is a tool for asymmetric-cost regimes, and we report the regime where it loses as plainly as the one where it wins.

11 Ablations and diagnostics

Unless stated, the ablations below were run on a single configuration, DLinear / ETTh1 / $H = 720$, under instant feedback, which is why their Proposed value (0.8571) differs from the 32-configuration mean in Table 1. [**TODO: re-run γ , scale and K on all 32 configurations under realised feedback; inject**]

Adaptation step size. Worst-cell coverage rises monotonically with γ (0.6520 at $\gamma = 0$, i.e. static conditional on this configuration; 0.7546 at 0.005; 0.8571 at 0.02; 0.8791 at 0.05; 0.8901 at 0.1) while width falls (3.5677 to 2.3627). We fixed $\gamma = 0.02$ before scoring and did not move it; the monotone trend indicates the result is not a knife-edge tuning artefact, and that larger γ would if anything have flattered us.

Scale estimator. With MAD normalisation the global baseline reaches 0.8549 worst-cell against 0.6869 with standard deviation, while Proposed is invariant at 0.8571: the conditional adaptive layer absorbs a choice that materially changes the baseline.

Stride. DLinear / ETTh1 / $H = 96$, the only setting with enough non-overlapping windows. Across strides 24 / 48 / 96 ($n_{\text{cal}} = 117/59/30$) Proposed scores 0.8702 / 0.8382 / 0.7500 and leads at every stride. The ordering is stable; the level falls with stride, which is consistent with the sample-size effect of Section 8.2 but cannot, at 30 non-overlapping paths, be separated from the optimism that overlapping windows can induce.

11.1 Is the gain just absorbing forecast bias?

BC-ACI [27] argues that a frozen forecaster develops persistent bias after a shift, and that symmetric intervals then pay a width overhead of roughly twice the bias. Our intervals are symmetric and our case-study split is seasonal, so the premise applies and we tested it.

The answer splits by backbone. DLinear carries per-cell bias of 16.5% of interval width on ETT, and it persists: calibration-block bias predicts test-block bias at $r = +0.579$ (ETT) and $+0.731$ (electricity), growing with horizon. NLinear carries 4.2% with no positive persistence ($r = -0.044$ on ETT and -0.595 on electricity, the latter an *anti*-persistence that a calibration-block correction would get wrong). The mechanism is architectural: NLinear subtracts the last observed value and re-anchors each window, while DLinear decomposes trend and seasonality and drifts.

Two conclusions follow. First, a leakage-free re-centring buys width, not coverage: pooled over ETT it returns -2.61% of width on DLinear and -0.44% on NLinear, and on the single worst DLinear configuration (ETTh2, $H = 720$) 11.2% of width while moving worst-cell coverage the wrong way (0.8547 to 0.8451). Bias correction is orthogonal to our objective and composable with our layer, since it acts on the score and we act on the quantile. Second, this rules out an alternative explanation for Section 7: if the horizon axis were merely soaking up horizon-dependent bias, its benefit would be far larger on DLinear, which carries about four times the bias as a fraction of width. It is not—+0.0406 on DLinear against +0.0383 on NLinear—and per configuration the association is slightly negative ($r = -0.317$). The horizon axis is capturing scale heterogeneity, consistent with the observation that per-horizon interval width is strongly daily-periodic (lag-24 autocorrelation +0.9810) rather than monotone in h ; a conditioning axis on hour-of-day rather than log-spaced horizon is the natural next ablation.

12 Limitations

Backbones. We evaluate two linear forecasters, not the Transformer architectures our study was designed around. The model-agnosticism claim is supported across two backbones with genuinely different error processes, but not across architectural families. The decisive test is PatchTST, which enforces channel independence by construction: if the channel axis is where calibration breaks, a channel-independent backbone should either shrink the gain or enlarge it, and neither result is available here.

Baselines. Conformal PID, AcMCP/MPID, CopulaCPTS and pinball-trained quantile heads were in our written comparison set and are absent from this draft. AcMCP is the dependence-aware, delayed-feedback method for exactly this problem and is the most important omission. **[TODO: add PID and AcMCP, or state the reason for their absence]**

Dependence and guarantees. Calibration windows overlap heavily, so cell counts are not sample sizes and the finite-sample split-conformal form is a convention. We claim long-run adaptive coverage and approximate bucketed conditioning, nothing stronger; exact conditional coverage is unattainable distribution-free [19], and our finite-sample marginal guarantee does not survive the non-exchangeability we deliberately evaluate under.

Feedback delay. Realised feedback costs the most at $H = 720$ (Section 6) and on electricity (Section 8); a sliding calibration pool that admits realised test scores, as in EnbPI [23], would reduce the cost and is not evaluated here.

Grid width and cross-surface comparison. Worst-cell coverage over 300 cells is not comparable to worst-cell coverage over 42; we report distributional statistics alongside and do not rank surfaces by their minima or by their interactions on minima.

Sample size and selection. On the 300-cell surface the method has not saturated in calibration data. The 50 meters are the first 50 zero-free columns, one split, one year of test data. All uncertainty statements are paired bootstraps over configurations, which share datasets and are not independent.

One reproduction discrepancy. DLinear at the longest horizons on ETTh2 and ETTm2 (Appendix B) remains above the published values; the closed-form fit is the likely cause.

13 Conclusion

Attaching conformal prediction to an LTSF forecaster is easy; attaching it so that coverage holds where it is needed is not. Scored marginally, seven interval methods are indistinguishable and the ranking recommends the one that fails hardest per cell. Scored conditionally, the obvious individual repairs do not help: adapting the quantile online reliably makes the worst cell worse, and conditioning it on horizon and channel helps or hurts depending on the dataset. What works is the combination, under an honest online protocol, at negligible computational cost and without touching the forecaster.

We think the transferable lessons are about evaluation. A single aggregate coverage number is compatible with systematic failure along horizon, channel and season; a minimum computed over a grid is not comparable across settings that change the grid; and an online update can pass a leakage test while consuming the future. All three bit us before they became findings, and all three are cheap to check.

Reproducibility. Code, all result artefacts, the generated number macros used to build this document, and the complete decision, failure and open-question registers—including the lost-work incidents, the scoring confound and the feedback leak described above—are released at <https://github.com/aarush093/coverage-across-the-horizon>. The main study runs in under four minutes on CPU.

A Instant-feedback (oracle) results

The numbers below are those of Draft v1. Each cell’s tracker consumed a path’s outcomes at all horizon steps before the next path was issued; they bound what realised feedback can achieve. Static rows are unchanged from the main text.

Surface	Method	Marginal	Worst-cell	5th pct. cell	Within ± 5	Width	Winkler
ETT	ACI (instant)	0.9031	0.7445	0.8100	0.213	1.8742	2.8310
ETT	Proposed (instant)	0.9098	0.8657	0.8711	0.888	1.9245	2.6366
Electricity	ACI (instant)	0.8921	0.6177	0.8353	0.488	0.9568	1.5152
Electricity	Proposed (instant)	0.8964	0.8306	0.8806	0.977	0.9420	1.4083

Interactions under instant feedback: ETT +0.1291 (CI (+0.093, +0.171)), electricity +0.3264. Per horizon on ETT, Proposed instant vs. realised: 0.8697 vs. 0.8690, 0.8661 vs. 0.8635, 0.8656 vs. 0.8562, 0.8616 vs. 0.8478. On electricity at $H = 720$: 0.8533 vs. 0.7748.

B Point-forecast sanity, all configurations

Backbone	Dataset	H	MSE	MAE	n_{cal}	n_{test}
DLinear	ETTh1	96	0.3702	0.3915	117	117
DLinear	ETTh1	192	0.4041	0.4127	113	113
DLinear	ETTh1	336	0.4334	0.4342	107	107
DLinear	ETTh1	720	0.4714	0.4878	91	91
DLinear	ETTh2	96	0.3018	0.3655	117	117
DLinear	ETTh2	192	0.3972	0.4276	113	113
DLinear	ETTh2	336	0.4858	0.4837	107	107
DLinear	ETTh2	720	0.7407	0.6116	91	91
DLinear	ETTm1	96	0.2993	0.3428	120	120
DLinear	ETTm1	192	0.3340	0.3637	119	119
DLinear	ETTm1	336	0.3684	0.3844	117	117
DLinear	ETTm1	720	0.4233	0.4181	113	113
DLinear	ETTm2	96	0.1723	0.2694	120	120
DLinear	ETTm2	192	0.2410	0.3228	119	119
DLinear	ETTm2	336	0.3189	0.3784	117	117
DLinear	ETTm2	720	0.4581	0.4633	113	113
NLinear	ETTh1	96	0.3702	0.3915	117	117
NLinear	ETTh1	192	0.4034	0.4112	113	113
NLinear	ETTh1	336	0.4277	0.4262	107	107
NLinear	ETTh1	720	0.4344	0.4505	91	91
NLinear	ETTh2	96	0.2726	0.3351	117	117
NLinear	ETTh2	192	0.3343	0.3762	113	113
NLinear	ETTh2	336	0.3588	0.4004	107	107
NLinear	ETTh2	720	0.3930	0.4346	91	91
NLinear	ETTm1	96	0.3005	0.3426	120	120
NLinear	ETTm1	192	0.3357	0.3634	119	119
NLinear	ETTm1	336	0.3704	0.3834	117	117
NLinear	ETTm1	720	0.4263	0.4157	113	113
NLinear	ETTm2	96	0.1647	0.2537	120	120
NLinear	ETTm2	192	0.2198	0.2910	119	119
NLinear	ETTm2	336	0.2734	0.3262	117	117
NLinear	ETTm2	720	0.3674	0.3831	113	113

[TODO: add the published column from Zeng et al. (2023), Table 2, look-back 336, and the percentage deviation]

C Bucket edges

$K = 6$ edges by the rule of Section 4: $H = 96$: [1, 2, 5, 10, 21, 45, 96]; $H = 192$: [TODO: print from buckets(192, 6)]; $H = 336$: [TODO: print]; $H = 720$: [1, 3, 9, 27, 80, 240, 720].

References

- [1] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, W. Zhang. Informer: Beyond efficient Transformer for long sequence time-series forecasting. *AAAI*, 2021.
- [2] A. Zeng, M. Chen, L. Zhang, Q. Xu. Are Transformers effective for time series forecasting? *AAAI*, 2023. (Introduces DLinear and NLinear.)

- [3] Y. Nie, N. H. Nguyen, P. Sinthong, J. Kalagnanam. A time series is worth 64 words: Long-term forecasting with Transformers. *ICLR*, 2023.
- [4] J. Zhang et al. ProbTS: Benchmarking point and distributional forecasting across diverse prediction horizons. *NeurIPS Datasets & Benchmarks*, 2024.
- [5] I. Gibbs, E. Candès. Adaptive conformal inference under distribution shift. *NeurIPS*, 2021.
- [6] I. Gibbs, E. Candès. Conformal inference for online prediction with arbitrary distribution shifts. *JMLR*, 25, 2024. arXiv:2208.08401.
- [7] M. Zaffran, O. Féron, Y. Goude, J. Josse, A. Dieuleveut. Adaptive conformal predictions for time series. *ICML*, 2022.
- [8] A. N. Angelopoulos, E. Candès, R. J. Tibshirani. Conformal PID control for time series prediction. *NeurIPS*, 2023.
- [9] X. Wang, R. J. Hyndman. Online conformal inference for multi-step time series forecasting. arXiv:2410.13115.
- [10] R. Li, D. Menacho, A. Rodríguez. Optimization-based online conformal prediction for multi-step forecasting. arXiv:2508.13362v2.
- [11] K. Stankeviciute, A. M. Alaa, M. van der Schaar. Conformal time-series forecasting. *NeurIPS*, 2021.
- [12] Z. Yang, E. Candès, L. Lei. Bellman conformal inference: Calibrating prediction intervals for time series. arXiv:2402.05203.
- [13] S. Sun, R. Yu. Copula conformal prediction for multi-step time series forecasting. *ICLR*, 2023. arXiv:2212.03281.
- [14] V. Vovk, D. Lindsay, I. Nourtdinov, A. Gammerman. Mondrian confidence machine. Technical report, Royal Holloway, University of London, 2003.
- [15] O. Bastani, V. Gupta, C. Jung, G. Noarov, R. Ramalingam, A. Roth. Practical adversarial multivalid conformal prediction. *NeurIPS*, 2022.
- [16] Y. Lai et al. Online localized conformal prediction. arXiv:2605.05497. **[TODO: complete author list]**
- [17] V. Vovk. Conditional validity of inductive conformal predictors. *ACML*, PMLR 25, 2012.
- [18] J. Lei, L. Wasserman. Distribution-free prediction bands for non-parametric regression. *JRSS-B*, 76(1), 2014.
- [19] R. Foygel Barber, E. J. Candès, A. Ramdas, R. J. Tibshirani. The limits of distribution-free conditional predictive inference. *Information and Inference*, 10(2), 2021.
- [20] I. Gibbs, J. J. Cherian, E. J. Candès. Conformal prediction with conditional guarantees. *JRSS-B*, 2025.
- [21] H. Papadopoulos, K. Proedrou, V. Vovk, A. Gammerman. Inductive confidence machines for regression. *ECML*, 2002.

- [22] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman. Distribution-free predictive inference for regression. *JASA*, 113(523), 2018.
- [23] C. Xu, Y. Xie. Conformal prediction interval for dynamic time-series. *ICML*, 2021.
- [24] R. L. Winkler. A decision-theoretic approach to interval estimation. *JASA*, 67(337), 1972.
- [25] X. Qiu et al. TFB: Towards comprehensive and fair benchmarking of time series forecasting methods. *PVLDB*, 17(9), 2024.
- [26] A. Sabashvili. Conformal prediction algorithms for time series forecasting: Methods and benchmarking. arXiv:2601.18509.
- [27] A. Lade, S. Krishna J., I. Kumar. Bias-corrected adaptive conformal inference for multi-horizon time series forecasting. arXiv:2604.13253.
- [28] K. Wood, S. Zohren, S. J. Roberts. DeRegiME: Deep regime mixtures for probabilistic forecasting under distribution shift. arXiv:2605.19231.
- [29] Z. Ma, J. Jiang, Á. López-Oriona, Y. Sun, H. Ombao. Adaptive multi-scale forecasting and gate-localized conformal prediction for multivariate nonstationary time series. arXiv:2607.23165.
- [30] Y. Liu et al. iTransformer: Inverted Transformers are effective for time series forecasting. *ICLR*, 2024.
- [31] Y. Wang et al. TimeXer: Empowering Transformers for time series forecasting with exogenous variables. *NeurIPS*, 2024.
- [32] A. Trindade. ElectricityLoadDiagrams20112014. UCI Machine Learning Repository, 2015.
- [33] G. Lai, W.-C. Chang, Y. Yang, H. Liu. Modeling long- and short-term temporal patterns with deep neural networks. *SIGIR*, 2018.